COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9/24/2026	**DATE SUBMITTED:**	9/24/2026
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MIDTERM **SKILLS EXAM**

1. PROCEDURE OUTPUT

Task 1: Prepare the Server

Updating the package repository for Ubuntu and CentOS

```
- name: TASK 1 update the repository index for UBUNTU
  ansible.builtin.apt:
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: TASK 1 update repository index for CENTOS
  ansible.builtin.dnf:
    update_cache: yes
  when: ansible_distribution == "CentOS"
```

Upgrading the installed packages for Ubuntu and CentOS

```
- name: TASK 1 install required utilities UBUNTU
  ansible.builtin.apt:
    name:
      - wget
      - curl
      - tar
      - gzip
    state: present
  when: ansible_distribution == "Ubuntu"
```

```
- name: TASK 1 install required utilities CENTOS
  ansible.builtin.dnf:
    name:
      - wget
      - curl
      - tar
      - gzip
    state: present
  when: ansible_distribution == "CentOS"
```

```
TASK [TASK 1 update the repository index for UBUNTU] *******
skipping: [192.168.56.116]
changed: [192.168.56.115]

TASK [TASK 1 update repository index for CENTOS] ***********
skipping: [192.168.56.115]
ok: [192.168.56.116]

TASK [TASK 1 install required utilities UBUNTU] ************
skipping: [192.168.56.116]
changed: [192.168.56.115]

TASK [TASK 1 install required utilities CENTOS] ************
skipping: [192.168.56.115]
ok: [192.168.56.116]
```

- We are updating as well as upgrading the packages and utilities so that there are no outdated versions of some packages which might result in incompatible versions when installing prometheus.

Task 2: Create the Prometheus User

```
- name: TASK 2 create prometheus system user
  ansible.builtin.user:
    name: prometheus
    system: yes
    shell: /bin/false
    create_home: no
```

```
TASK [TASK 2 create prometheus system user] ******
changed: [192.168.56.115]
changed: [192.168.56.116]
```

- Creating a prometheus system user so that we can use prometheus once it is installed and it isolates the monitoring system from the operating system itself.

Task 3: Create Prometheus Directories

```
- name: TASK 3 create the prometheus directories
  ansible.builtin.file:
    path: "{{ item }}"
    state: directory
    owner: prometheus
    group: prometheus
    mode: '0755'
  loop:
    - /etc/prometheus
    - /var/lib/prometheus
```

```
TASK [TASK 3 create the prometheus directories] **********
changed: [192.168.56.115] => (item=/etc/prometheus)
changed: [192.168.56.115] => (item=/var/lib/prometheus)
changed: [192.168.56.116] => (item=/etc/prometheus)
changed: [192.168.56.116] => (item=/var/lib/prometheus)
```

- This part is about creating the prometheus directories as well as assigning the owner and group.

Task 4: Download Prometheus

```
- name: TASK 4 which is downloading the epic prometheus
  ansible.builtin.unarchive:
    src: "https://github.com/prometheus/prometheus/releases/download/v3.14.0/prometheus-3.14.0.linux-amd64.tar.gz"
    dest: /tmp
    remote_src: yes
```

```
TASK [TASK 4 which is downloading the epic prometheus]
changed: [192.168.56.116]
changed: [192.168.56.115]
```

- This is simply just to download the prometheus along with the current version we want to put.

Task 5: Install the Prometheus Binaries

```
- name: TASK 5 which is to copy binaries to /usr/local/bin
  ansible.builtin.copy:
    src: "/tmp/prometheus-3.14.0.linux-amd64/{{ item }}"
    dest: "/usr/local/bin/{{ item }}"
    owner: prometheus
    group: prometheus
    mode: '0755'
    remote_src: yes
  loop:
    - prometheus
    - promtool
```

```
TASK [TASK 5 which is to copy binaries to /usr/local/bin] *
ok: [192.168.56.115] => (item=prometheus)
ok: [192.168.56.116] => (item=prometheus)
ok: [192.168.56.115] => (item=promtool)
ok: [192.168.56.116] => (item=promtool)
```

- This part was for installing the prometheus binaries but all I had to do was copy them to their designated location with the use of an ansible playbook.

Task 6: Install the Prometheus Configuration Files

```
- name: TASK 6 aka copying the default prometheus.yml to the /etc/prometheus location
  ansible.builtin.copy:
    src: "/tmp/prometheus-3.14.0.linux-amd64/prometheus.yml"
    dest: /etc/prometheus/prometheus.yml
    owner: prometheus
    group: prometheus
    mode: '0644'
  remote_src: yes
```

```
TASK [TASK 6 aka copying the default prometheus.yml to the /etc/prometheus location] *
ok: [192.168.56.115]
changed: [192.168.56.116]
```

- This was to move the prometheus.yml to /etc/promethus location since the prometheus.yml are the configuration files.

Task 7: Validate the Configuration

Task 8: Create the systemd Service

```
- name: TASK 8 which is to create systemd service file for Prometheus
  ansible.builtin.copy:
    dest: /etc/systemd/system/prometheus.service
    content: |
      [Unit]
      Description=Prometheus Time Series Collection and Processing Server
      Wants=network-online.target
      After=network-online.target

      [Service]
      User=prometheus
      Group=prometheus
      Type=simple
      ExecStart=/usr/local/bin/prometheus \
        --config.file=/etc/prometheus/prometheus.yml \
        --storage.tsdb.path=/var/lib/prometheus
      Restart=always

      [Install]
      WantedBy=multi-user.target
    mode: '0644'
```

```
TASK [TASK 8 which is to create systemd service file for Prometheus]
changed: [192.168.56.115]
changed: [192.168.56.116]
```

- Here, I simply added the systemd service from the seatwork while modifying some of the content.

Task 9: Start and Enable Prometheus

```
  #this is task 9
- name: TASK 9 which is to start then enable then reload prometheus
  ansible.builtin.systemd:
    name: prometheus
    state: started
    enabled: yes
    daemon_reload: yes
```

```
TASK [TASK 9 which is to start then enable then reload prometheus] *
changed: [192.168.56.116]
changed: [192.168.56.115]
```

- Here, I simply just started, enabled then reloaded prometheus which we can see that it worked properly based on the output.

Task 10: Verify Prometheus

Server1 (Ubuntu)

```
momo@server1:~/Desktop$ prometheus --version
prometheus, version 3.14.0 (branch: HEAD, revision: d7598b7141418fa35be2b5ec5d0f
efb634199610)
  build user:       root@f423027f4410
  build date:       20260817-16:49:19
  go version:       go1.26.6
  platform:         linux/amd64
  tags:             netgo,builtinassets
momo@server1:~/Desktop$
```

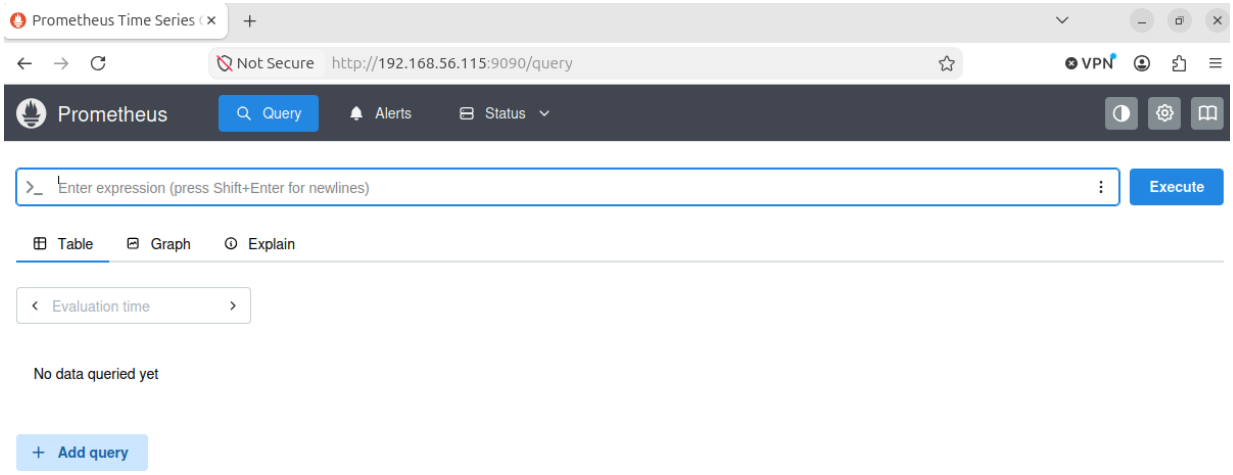
Server2 (CentOS)

```
[momo@centos ~]$ prometheus --version
prometheus, version 3.14.0 (branch: HEAD, revision: d7598b7141418fa35be2b5ec5d0
efb634199610)
  build user:       root@f423027f4410
  build date:       20260817-16:49:19
  go version:       go1.26.6
  platform:         linux/amd64
  tags:             netgo,builtinassets
[momo@centos ~]$
```

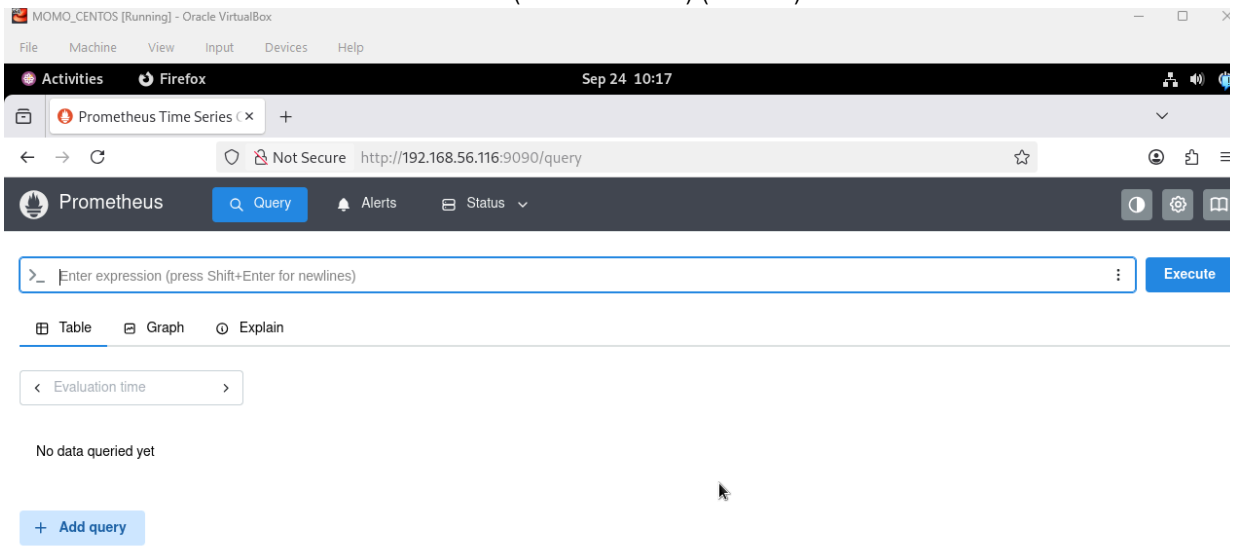
- We can see that using the command prometheus –version, it was successfully installed.

Task 11: Access Prometheus Through a Browser

Server1 (192.168.56.115) (Ubuntu)



Server2 (192.168.56.116) (CentOS)



- We can see that inserting the IP address along with the ":9090", it brought us to the prometheus time series and collection server which indicates that it worked properly! :D

WHOLE PLAYBOOK:

```
momo@workstation:~/CPE212_MOMO/midtermskills$ cat install_prometheus.yml
---
- hosts: all
  become: true
  tasks:

    #this is all task 1
  - name: TASK 1 update the repository index for UBUNTU
    ansible.builtin.apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: TASK 1 update repository index for CENTOS
    ansible.builtin.dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: TASK 1 install required utilities UBUNTU
    ansible.builtin.apt:
      name:
        - wget
        - curl
        - tar
        - gzip
      state: present
    when: ansible_distribution == "Ubuntu"
```

```
  - name: TASK 1 install required utilities CENTOS
    ansible.builtin.dnf:
      name:
        - wget
        - curl
        - tar
        - gzip
      state: present
    when: ansible_distribution == "CentOS"

    #this is task 2
  - name: TASK 2 create prometheus system user
    ansible.builtin.user:
      name: prometheus
      system: yes
      shell: /bin/false
      create_home: no

    #this is task 3
  - name: TASK 3 create the prometheus directories
    ansible.builtin.file:
      path: "{{ item }}"
      state: directory
      owner: prometheus
      group: prometheus
      mode: '0755'
    loop:
      - /etc/prometheus
      - /var/lib/prometheus
```

```
- name: TASK 4 which is downloading the epic prometheus
  ansible.builtin.unarchive:
    src: "https://github.com/prometheus/prometheus/releases/download/v3.14.0/prometheus-3.14.0.linux-amd64.tar.gz
    dest: /tmp
    remote_src: yes

  #this is task 5
- name: TASK 5 which is to copy binaries to /usr/local/bin
  ansible.builtin.copy:
    src: "/tmp/prometheus-3.14.0.linux-amd64/{{ item }}"
    dest: "/usr/local/bin/{{ item }}"
    owner: prometheus
    group: prometheus
    mode: '0755'
    remote_src: yes
  loop:
    - prometheus
    - promtool

  #this is task 6
- name: TASK 6 aka copying the default prometheus.yml to the /etc/prometheus location
  ansible.builtin.copy:
    src: "/tmp/prometheus-3.14.0.linux-amd64/prometheus.yml"
    dest: /etc/prometheus/prometheus.yml
    owner: prometheus
    group: prometheus
    mode: '0644'
    remote_src: yes

  #this is task 7
```

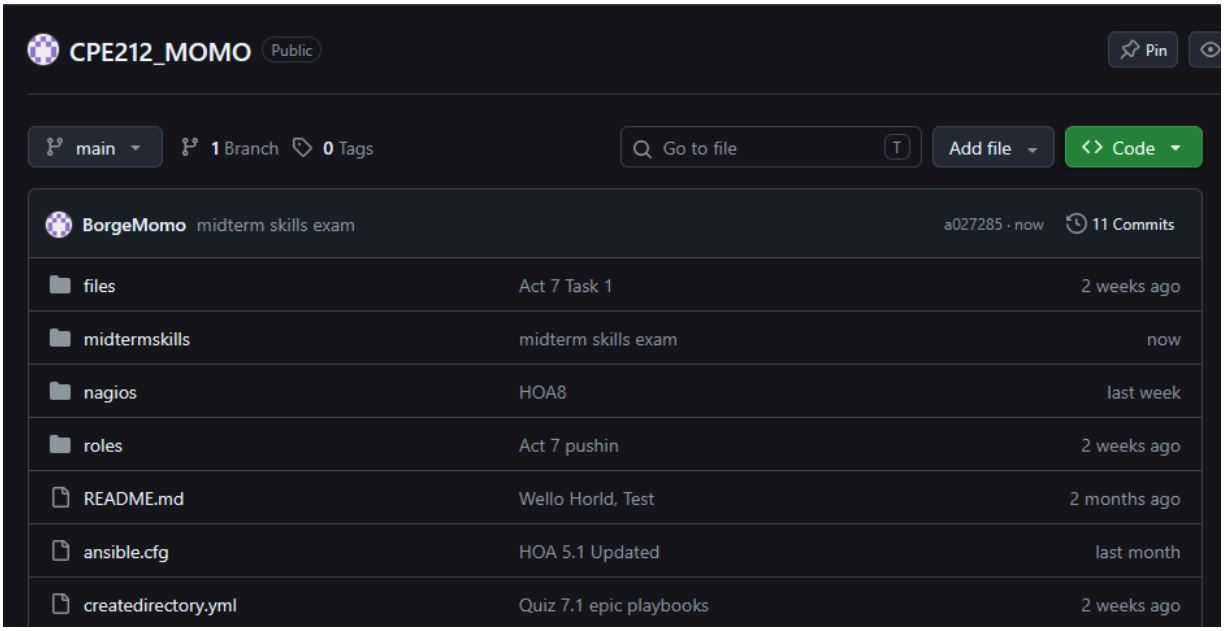
```
  #this is task 8
- name: TASK 8 which is to create systemd service file for Prometheus
  ansible.builtin.copy:
    dest: /etc/systemd/system/prometheus.service
    content: |
      [Unit]
      Description=Prometheus Time Series Collection and Processing Server
      Wants=network-online.target
      After=network-online.target

      [Service]
      User=prometheus
      Group=prometheus
      Type=simple
      ExecStart=/usr/local/bin/prometheus \
        --config.file=/etc/prometheus/prometheus.yml \
        --storage.tsdb.path=/var/lib/prometheus
      Restart=always

      [Install]
      WantedBy=multi-user.target
    mode: '0644'

  #this is task 9
- name: TASK 9 which is to start then enable then reload prometheus
  ansible.builtin.systemd:
    name: prometheus
    state: started
    enabled: yes
    daemon_reload: yes
```

Github PUSH verification: (it was still inside CPE212_MOMO)



https://github.com/BorgeMomo/CPE212_MOMO

CONCLUSION

In conclusion, this midterm skills exam was pretty hard at first but analyzing the different hands on activities as well as the seatwork from last meeting, I finished this midterm skills exam, it was pretty hard at first but figuring things out along the way, I managed to finish it but I missed task 7 because it was pretty confusing.

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